

Integral and Transform Calculus for the Fabius–Rvachev–Quantile System

Order-statistic spacings, beta–quantile lattices, dyadic resolvents,
sinc energies, and beyond-all-orders localization

Research report prepared for Vladimir Reshetnikov

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Abstract

This report develops a unified calculus for indefinite and definite integrals, iterated and fractional antiderivatives, and Mellin, Laplace, Fourier, Cauchy–Stieltjes, and beta transforms involving the bounded Fabius function F , Rvachev’s compactly supported function u , and the inverse Fabius function $G = F^{-1}$. The repository corpus was audited at the pinned snapshot a6555a64671b; its recent monomial/fractional-calculus, inverse-asymptotic, Fourier-decay, and Bell–Bernoulli developments are treated as baseline rather than rediscovered.

The main new organizing principle is an adjacent-order-statistic spacing identity. It converts every weighted mixed integral $\int h(x)F(x)^p(1 - F(x))^q dx$ into the expected increment of an antiderivative of h across two consecutive order statistics. This yields, in one stroke, explicit formulas for polynomial, Jacobi, exponential, trigonometric, logarithmic, Lambert- W , Mellin, Fourier, and generalized resolvent weights, as well as a closed probabilistic form for every integer or fractional Volterra primitive. A complementary beta–quantile calculus gives incomplete antiderivatives in terms of truncated moments of G , a Pascal lattice, reflection identities, and strict checkerboard total positivity.

Further results include: a monotone beta-normalized balanced sequence converging to $1/2$ faster than every inverse power; its logarithmic decay constant $-1/(8 \log 2)$, with a Lambert- W saddle; a Cauchy–Stieltjes hierarchy for the up-law satisfying exact dyadic functional-differential equations; a CDF/quantile resolvent duality; endpoint derivative growth of order $\exp((\log 2)n^2/2)$; and an exact complementary pair of sinc-product energy integrals for the previously opaque constant $\int_0^1 F(x)^2 dx$. A commented Python program independently checks the principal identities and is included with the source and PDF.

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1 Scope, corpus audit, and status of the results

1.1 Audit boundary

The audit covered the `*.tex` sources under

`Analysis/FabiusFunction/docs`

including all subdirectories. The working snapshot was pinned to commit `a6555a64671b`, after the merge of the recent `frontier-report` stack. The pinned Git-tree enumeration used in the audit contained 87 TeX paths; its complete path manifest is included with this report. Byte-identical copies, source/PDF bundles, OCR transcriptions of external papers, and successive draft variants were grouped by content lineage; theorem-level comparison was performed against every content-distinct Fabius/Rvachev source and against the archive ledger.

The closest predecessors are the reports on monomial antiderivatives, complex-power Newton–Volterra expansions, fractional calculus, inverse-Fabius asymptotics, derivative norm spectra, and the consolidated frontier reports. Consequently, the following are treated here as *baseline*:

- the primitive ladder for $x^\alpha F(x)$, including the terminating integer-power case and convergent complex-power series;
- basic Volterra/Riemann–Liouville formulas for F and G ;
- endpoint asymptotics expressed through Lambert W_{-1} , periodic corrections, Bell polynomials, and Gamma–zeta operators;
- the infinite sinc product for $\uparrow p$, its dyadic refinement, and the existing Fourier-decay analysis;
- exact dyadic values, rational moment recurrences, finite Thue–Morse splines, and the established q -binomial machinery.

A superscript ^N marks a result that appears new relative to the audited repository corpus; ^B marks a recovered or reorganized baseline fact; and ^C marks a conjecture. These labels are not claims of global priority in the mathematical literature.

1.2 Principal new contributions

The report’s main contributions are:

1. the weighted adjacent-spacing theorem and its partial, iterated, and fractional antiderivative forms;
2. an incomplete beta–quantile antiderivative and a two-parameter beta–quantile lattice;
3. a bivariate resolvent for all mixed powers, with Möbius symmetry and strict checkerboard total positivity;
4. a beta-localization theorem, strict monotonicity of the balanced normalized sequence, superalgebraic convergence, and the exact leading logarithmic rate governed by a Lambert saddle;
5. a Cauchy–Stieltjes hierarchy for $\uparrow p$, including boundary-value and endpoint nonanalyticity statements;
6. a generalized resolvent duality interchanging F and G ;
7. the sinc-energy identity and complementary Gini sum rule for $\int_0^1 F^2$;
8. a certified rational finite-spline approximation scheme inherited from the Thue–Morse truncations, with an explicit coupling error bound.

2 The three functions and their probability models

2.1 The bounded Fabius distribution

Let $(U_j)_{j \geq 1}$ be independent $\text{Unif}[0, 1]$ variables and set

$$X = \sum_{j \geq 1} \frac{U_j}{2^j}. \quad (1)$$

The bounded Fabius function is the distribution function

$$F(x) = \mathbb{P}(X \leq x), \quad 0 \leq x \leq 1, \quad (2)$$

continued by 0 to the left and 1 to the right whenever a genuine CDF is needed. It obeys

$$F(1-x) = 1 - F(x), \quad F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad (3)$$

and is strictly increasing on $[0, 1]$. Its inverse is

$$G(u) = F^{-1}(u), \quad 0 \leq u \leq 1, \quad (4)$$

with $G(1-u) = 1 - G(u)$.

Write

$$d_s = \mathbb{E}X^s = \int_0^1 x^s F'(x) dx \quad (5)$$

whenever the moment exists. For integer $n \geq 1$, the exact recurrence is

$$(n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k. \quad (6)$$

In particular,

$$d_0 = 1, \quad d_1 = \frac{1}{2}, \quad d_2 = \frac{5}{18}, \quad d_3 = \frac{1}{6}, \quad d_4 = \frac{143}{1350}.$$

2.2 Rvachev's compactly supported density

Put

$$Y = 2X - 1 = \sum_{j \geq 1} \frac{V_j}{2^j}, \quad V_j = 2U_j - 1 \sim \text{Unif}[-1, 1]. \quad (7)$$

Then Y has density

$$\text{up}(y) = F(1 - |y|) \mathbf{1}_{[-1, 1]}(y), \quad F'(x) = 2 \text{up}(2x - 1). \quad (8)$$

Thus up is even and C^∞ , and

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1)). \quad (9)$$

We use the cycles-per-unit Fourier convention

$$\hat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad \text{sinc } t = \frac{\sin t}{t}. \quad (10)$$

Then

$$\Phi(\xi) := \mathfrak{up}(\xi) = \prod_{j \geq 0} \operatorname{sinc}\left(\frac{\pi \xi}{2^j}\right). \quad (11)$$

The moment-generating functions are

$$M_X(z) = \mathbb{E}e^{zX} = \prod_{j \geq 1} \frac{e^{z/2^j} - 1}{z/2^j}, \quad (12)$$

$$M_Y(z) = \mathbb{E}e^{zY} = \prod_{j \geq 1} \frac{\sinh(z/2^j)}{z/2^j}. \quad (13)$$

2.3 Bernoulli cumulants and Bell moments

The transform (13) makes the Bell–Bernoulli structure transparent. Since

$$\log \frac{\sinh z}{z} = \sum_{m \geq 1} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m},$$

one obtains

$$\log M_Y(z) = \sum_{m \geq 1} \frac{2^{2m-1} B_{2m}}{m(2m)!(2^{2m} - 1)} z^{2m}. \quad (14)$$

Hence the nonzero cumulants are

$$\kappa_{2m}(Y) = \frac{2^{2m-1} B_{2m}}{m(2^{2m} - 1)}, \quad \kappa_{2m+1}(Y) = 0, \quad (15)$$

and the centered moments $c_m = \mathbb{E}Y^{2m}$ are complete Bell polynomials in these cumulants. The first values are

$$c_0 = 1, \quad c_1 = \frac{1}{9}, \quad c_2 = \frac{19}{675}, \quad c_3 = \frac{583}{59535}.$$

They also satisfy $c_m = 2d_{2m+1}/(2m+1)$.

3 Endpoint integrability: powers of a CDF versus powers of a quantile

The endpoint law established in the corpus has the logarithmic form

$$\log F(e^{-t}) = -\frac{t^2}{2 \log 2} + O(t \log t), \quad t \rightarrow +\infty, \quad (16)$$

with much sharper Lambert-periodic expansions available in the baseline reports. This immediately creates a useful dichotomy.

Proposition 3.1 (Integrability classification). *Let $p, q, \alpha, \beta \in \mathbb{C}$.*

- (i) *The mixed power $F(x)^p(1-F(x))^q$ is absolutely integrable on $(0, 1)$ precisely when $\Re p \geq 0$ and $\Re q \geq 0$.*
- (ii) *If $\Re p > 0$, then $x^\alpha F(x)^p$ is integrable at 0 for every α . If $\Re p = 0$, one needs $\Re \alpha > -1$; if $\Re p < 0$, no algebraic weight repairs the divergence.*

(iii) Every fixed negative moment of X is finite:

$$\mathbb{E}X^{-s} = \int_0^1 G(u)^{-s} du < \infty \quad (s > 0),$$

whereas $\int_0^1 F(x)^{-s} dx = \infty$.

(iv) For $\Re \alpha > -1$, the integral $\int_{-1}^1 |x|^\alpha \text{up}(x)^q dx$ converges when $\Re q \geq 0$ and diverges when $\Re q < 0$.

Proof. Put $x = e^{-t}$. By (16), a negative power $F(x)^{-s}$ contributes $\exp(st^2/(2 \log 2) - t + O(t \log t)) dt$, which diverges. A positive power has the opposite quadratic sign and dominates every algebraic singularity. For the quantile, inversion of (16) gives $G(e^{-L}) = \exp(-\sqrt{2 \log 2} L + O(\log L))$. Thus $G(u)^{-s} du$, under $u = e^{-L}$, has exponent $-L + O(\sqrt{L})$, and is integrable. The remaining endpoint follows from symmetry, while $\text{up}(1-x) = F(x)$ on $[0, 1]$. $\square$

This contrast explains why negative powers of F or up generally do not possess ordinary definite integrals, while inverse-Fabius negative moments are perfectly well behaved.

4 The adjacent-order-statistic spacing principle

4.1 The universal weighted identity

Let $Z_1, \dots, Z_N$ be independent copies of X , and write $Z_{1:N} \leq \dots \leq Z_{N:N}$ for their order statistics.

Theorem 4.1 (Weighted adjacent-spacing theorem^N). *Let $p, q \in \mathbb{N}_{>0}$, $N = p + q$, and let $h \in L^1(0, 1)$ have an absolutely continuous primitive H , so $H' = h$ almost everywhere. Then*

$$\int_0^1 h(x) F(x)^p (1 - F(x))^q dx = \frac{\mathbb{E}[H(Z_{p+1:N}) - H(Z_{p:N})]}{\binom{N}{p}}. \quad (17)$$

More generally, for $0 \leq a < b \leq 1$,

$$\int_a^b h(x) F(x)^p (1 - F(x))^q dx = \mathbb{E} \left[\mathbf{1}_{\{L < U\}} \int_a^b h(x) \mathbf{1}_{[L, U)}(x) dx \right], \quad (18)$$

where L is the maximum of p independent copies of X , and U is the minimum of another q copies.

Proof. For every fixed x , independence gives

$$\mathbb{P}(L \leq x < U) = F(x)^p (1 - F(x))^q.$$

Fubini's theorem yields (18). On $\{L < U\}$, all p observations in the first group precede all q observations in the second. Conditional on the unlabeled ordered sample, the labels are uniformly distributed among the $\binom{N}{p}$ possibilities. Exactly one labeling has the first p observations in the lower group. On that labeling, $L = Z_{p:N}$ and $U = Z_{p+1:N}$. Integrating h across the random gap gives (17). $\square$

The theorem is distribution-free: the Fabius law enters through the order statistics, not through the proof. Its value here is that the order statistics inherit the exact dyadic, inverse-quantile, and endpoint structures of F .

4.2 Moments, Jacobi weights, exponentials, and Lambert weights

Taking $H(x) = x^{m+1}/(m+1)$ gives

$$\int_0^1 x^m F(x)^p (1 - F(x))^q dx = \frac{\mathbb{E}[Z_{p+1:N}^{m+1} - Z_{p:N}^{m+1}]}{(m+1)\binom{N}{p}}. \quad (19)$$

For Jacobi weights, with the incomplete beta function $B_x(a, b) = \int_0^x t^{a-1}(1-t)^{b-1} dt$,

$$\int_0^1 x^\alpha (1-x)^\beta F(x)^p (1-F(x))^q dx = \frac{\mathbb{E}[B_{Z_{p+1:N}}(\alpha+1, \beta+1) - B_{Z_{p:N}}(\alpha+1, \beta+1)]}{\binom{N}{p}}. \quad (20)$$

Differentiating in α or β supplies all logarithmic Jacobi weights.

For $z \neq 0$,

$$\int_0^1 e^{zx} F(x)^p (1-F(x))^q dx = \frac{\mathbb{E}(e^{zZ_{p+1:N}} - e^{zZ_{p:N}})}{z\binom{N}{p}}, \quad (21)$$

$$\int_0^1 e^{-2\pi i \xi x} F(x)^p (1-F(x))^q dx = \frac{\mathbb{E}(e^{-2\pi i \xi Z_{p:N}} - e^{-2\pi i \xi Z_{p+1:N}})}{2\pi i \xi \binom{N}{p}}. \quad (22)$$

The limits at $z = 0$ or $\xi = 0$ recover the mean spacing.

A Lambert- W weight also has an elementary primitive:

$$\frac{d}{dx} \left(W(cx) + \frac{1}{2} W(cx)^2 \right) = \frac{W(cx)}{x}. \quad (23)$$

Therefore, on any branch for which the path avoids the branch cut,

$$\int_0^1 \frac{W(cx)}{x} F(x)^p (1-F(x))^q dx = \frac{\mathbb{E}[\mathcal{W}_c(Z_{p+1:N}) - \mathcal{W}_c(Z_{p:N})]}{\binom{N}{p}}, \quad (24)$$

where $\mathcal{W}_c(x) = W(cx) + \frac{1}{2} W(cx)^2$. This is a direct integral appearance of Lambert W , distinct from its asymptotic role at the endpoint.

4.3 Partial, iterated, and fractional antiderivatives

For $x \in [0, 1]$, the partial primitive has the exact random-interval form

$$\int_0^x h(t) F(t)^p (1-F(t))^q dt = \mathbb{E}[\mathbf{1}_{\{L < x, L < U\}} (H(\min\{x, U\}) - H(L))]. \quad (25)$$

This is already an indefinite integral: changing the lower endpoint adds only a constant.

The formula becomes especially sharp for the Riemann–Liouville operator

$$(I^\nu f)(x) = \frac{1}{\Gamma(\nu)} \int_0^x (x-t)^{\nu-1} f(t) dt, \quad \Re \nu > 0. \quad (26)$$

Theorem 4.2 (Fractional adjacent-spacing primitive^N). *For $p, q \in \mathbb{N}_{>0}$, $N = p + q$, $x \in [0, 1]$, and $\Re \nu > 0$,*

$$I^\nu [F^p (1-F)^q](x) = \frac{\mathbb{E}[(x - Z_{p:N})_+^\nu - (x - Z_{p+1:N})_+^\nu]}{\Gamma(\nu+1)\binom{N}{p}}. \quad (27)$$

For integer $n \geq 1$, the right side is the normalized difference of two truncated n -th moments and is the unique n -fold primitive whose first n initial data vanish at 0.

Proof. Insert $F(t)^p(1 - F(t))^q = \mathbb{P}(L \leq t < U)$ into the Volterra integral and apply Fubini. For fixed $L < U$,

$$\frac{1}{\Gamma(\nu)} \int_0^x (x - t)^{\nu-1} \mathbf{1}_{[L,U)}(t) dt = \frac{(x - L)_+^\nu - (x - U)_+^\nu}{\Gamma(\nu + 1)}.$$

The same random-label argument as in theorem 4.1 supplies the factor $\binom{N}{p}^{-1}$. $\square$

Endpoint cases are also useful. If $M_p = Z_{p:p}$ and $m_q = Z_{1:q}$, then

$$I^\nu[F^p](x) = \frac{\mathbb{E}(x - M_p)_+^\nu}{\Gamma(\nu + 1)}, \quad (28)$$

$$I^\nu[(1 - F)^q](x) = \frac{x^\nu - \mathbb{E}(x - m_q)_+^\nu}{\Gamma(\nu + 1)}. \quad (29)$$

For $p = 1$, (28) collapses to the baseline Fabius ladder

$$\mathbb{E}(x - X)_+^n = n! 2^{\binom{n}{2}} F(x/2^n). \quad (30)$$

Thus theorem 4.2 is a genuine mixed-power extension of that ladder.

5 Incomplete beta–quantile antiderivatives

5.1 A closed incomplete formula

For $r \in \mathbb{C}$, $a, b > 0$, and $0 \leq y \leq 1$, define

$$Q_r(a, b; y) = \int_0^y G(u)^r u^{a-1} (1 - u)^{b-1} du, \quad Q_r(a, b) = Q_r(a, b; 1). \quad (31)$$

Theorem 5.1 (Incomplete beta–quantile primitive^N). *Let $m > -1$, $p, q > 0$, $a = F(x)$. Whenever the displayed integrals converge,*

$$\begin{aligned} \int_0^x t^m F(t)^p (1 - F(t))^q dt &= \frac{x^{m+1} a^p (1 - a)^q}{m + 1} \\ &\quad - \frac{p}{m + 1} Q_{m+1}(p, q + 1; a) + \frac{q}{m + 1} Q_{m+1}(p + 1, q; a). \end{aligned} \quad (32)$$

Consequently,

$$\int_0^1 t^m F(t)^p (1 - F(t))^q dt = \frac{q Q_{m+1}(p + 1, q) - p Q_{m+1}(p, q + 1)}{m + 1}. \quad (33)$$

Proof. Integrate by parts with $t^m dt = d(t^{m+1}/(m + 1))$. The derivative of $F^p(1 - F)^q$ is

$$[p F^{p-1} (1 - F)^q - q F^p (1 - F)^{q-1}] F'.$$

Substitute $u = F(t)$ in the two resulting terms. The endpoint term vanishes at $x = 1$ when $q > 0$. $\square$

Formula (32) is a deterministic counterpart of the random-gap formula (25). It is often better for symbolic work because all dependence on the inverse Fabius function is isolated in Q_r .

5.2 The beta-quantile lattice

The complete transforms obey exact algebraic identities.

Proposition 5.2 (Pascal, reflection, and order-statistic laws^N). *For admissible parameters,*

$$Q_r(a, b) = Q_r(a + 1, b) + Q_r(a, b + 1), \quad (34)$$

$$Q_1(a, b) + Q_1(b, a) = B(a, b), \quad (35)$$

$$Q_r(b, a) = \sum_{k=0}^r (-1)^k \binom{r}{k} Q_k(a, b) \quad (r \in \mathbb{N}). \quad (36)$$

If $a, b \in \mathbb{N}_{>0}$, then

$$Q_r(a, b) = B(a, b) \mathbb{E} Z_{a: a+b-1}^r. \quad (37)$$

Proof. Equation (34) is the identity $u^{a-1}(1-u)^{b-1} = u^a(1-u)^{b-1} + u^{a-1}(1-u)^b$. For reflection, substitute $u \mapsto 1-u$ and use $G(1-u) = 1-G(u)$. The same substitution followed by the binomial theorem gives (36). Finally, $F(Z_{a: a+b-1}) \sim \text{Beta}(a, b)$, proving (37). $\square$

For $q_s = Q_1(s, 1)$, integration by parts gives

$$q_s = \int_0^1 u^{s-1} G(u) du = \frac{1 - A_s}{s}, \quad A_s := \int_0^1 F(x)^s dx. \quad (38)$$

When $b \in \mathbb{N}$,

$$Q_1(a, b) = \sum_{j=0}^{b-1} (-1)^j \binom{b-1}{j} q_{a+j}. \quad (39)$$

Thus the entire integer beta-quantile lattice is a finite binomial transform of the one-dimensional sequence (A_s) .

5.3 Unaligned products of F and G

Compositions aligned with the inverse are immediate:

$$G(F(x)) = x, \quad F(G(u)) = u.$$

Products at the same external argument, such as $F(x)^a G(x)^b$, do not collapse. They nevertheless admit an exact two-dimensional layer-cake kernel. For $a, b > 0$,

$$\int_0^1 F(x)^a G(x)^b dx = ab \int_0^1 \int_0^1 u^{a-1} v^{b-1} [1 - \max\{G(u), F(v)\}] du dv. \quad (40)$$

Indeed, $F(x)^a = a \int_0^1 u^{a-1} \mathbf{1}_{\{G(u) \leq x\}} du$, while $G(x)^b = b \int_0^1 v^{b-1} \mathbf{1}_{\{F(v) \leq x\}} dv$. Formula (40) is a promising starting point for genuinely nonlinear F - G coupling constants.

6 The mixed-power lattice

Define

$$I_{p,q} = \int_0^1 F(x)^p (1 - F(x))^q dx, \quad p, q \geq 0. \quad (41)$$

For integer arguments, the elementary partition $1 = F + (1 - F)$ gives

$$I_{p,q} = I_{p+1,q} + I_{p,q+1}, \quad I_{p,q} = I_{q,p}. \quad (42)$$

If $\Delta A_p = A_{p+1} - A_p$, then

$$I_{p,q} = (-1)^q \Delta^q A_p = \sum_{j=0}^q (-1)^j \binom{q}{j} A_{p+j}. \quad (43)$$

For noninteger q , an Abel-regularized Newton series remains valid:

$$I_{p,q} = \lim_{r \uparrow 1} \sum_{j \geq 0} (-1)^j \binom{q}{j} r^j A_{p+j}, \quad \Re p, \Re q \geq 0. \quad (44)$$

6.1 The bivariate resolvent

Let

$$R(z) = \int_0^1 \frac{dx}{1 - zF(x)} = \sum_{p \geq 0} A_p z^p, \quad |z| < 1. \quad (45)$$

Theorem 6.1 (Bivariate mixed-power generating function^N). *For $|z|, |w| < 1$, with removable points understood by analytic continuation,*

$$\boxed{\sum_{p,q \geq 0} I_{p,q} z^p w^q = \frac{zR(z) + wR(w)}{z + w - zw}}. \quad (46)$$

Moreover,

$$R(z) = \frac{1}{1-z} R\left(-\frac{z}{1-z}\right). \quad (47)$$

Proof. Sum the geometric series under the integral:

$$\sum_{p,q \geq 0} I_{p,q} z^p w^q = \int_0^1 \frac{dx}{(1 - zF(x))(1 - w(1 - F(x)))}.$$

Partial fractions give coefficients $z/(z + w - zw)$ and $w/(z + w - zw)$. Reflection $x \mapsto 1 - x$ changes the second integral into $R(w)$. The same reflection applied directly to R gives (47). $\square$

Parameter differentiation supplies mixed logarithmic integrals:

$$\frac{\partial^{r+s}}{\partial p^r \partial q^s} I_{p,q} = \int_0^1 F(x)^p (1 - F(x))^q (\log F(x))^r (\log(1 - F(x)))^s dx. \quad (48)$$

6.2 Strict checkerboard total positivity

The lattice has much stronger structure than Pascal recursion.

Theorem 6.2 (Strict sign-regularity^N). *Let $p_1 < \dots < p_m$ and $q_1 < \dots < q_m$ be nonnegative integers. Then*

$$(-1)^{\binom{m}{2}} \det [I_{p_i, q_j}]_{i,j=1}^m > 0. \quad (49)$$

Equivalently, reversing the order of the q -columns produces a strictly totally positive matrix.

Proof. Under $u = F(x)$,

$$I_{p,q} = \int_0^1 u^p (1-u)^q G'(u) du.$$

The continuous Cauchy–Binet (Andreief) identity gives

$$\det[I_{p_i, q_j}] = \frac{1}{m!} \int_{(0,1)^m} \det[u_k^{p_i}]_{i,k} \det[(1-u_k)^{q_j}]_{j,k} \prod_{k=1}^m G'(u_k) du_k.$$

On $0 < u_1 < \dots < u_m < 1$, the first generalized Vandermonde determinant is positive. The numbers $1 - u_k$ occur in decreasing order, so the second has sign $(-1)^{\binom{m}{2}}$. Both are nonzero away from diagonals, and $G' > 0$, proving strictness. $\square$

For $m = 2$, this yields the sharp reverse-TP inequality

$$I_{p,q} I_{p+1,q+1} < I_{p,q+1} I_{p+1,q}. \quad (50)$$

For fixed q , the sequence $p \mapsto I_{p,q}$ is a strict Hausdorff moment sequence and hence has positive Hankel minors; theorem 6.2 adds the cross-parameter sign law.

7 Beta localization and asymptotics

7.1 Interior localization

Substitution $u = F(x)$ gives, for a weight h ,

$$\int_0^1 h(x) F(x)^p (1-F(x))^q dx = B(p+1, q+1) \mathbb{E}K(B_{p,q}), \quad (51)$$

where

$$K(u) = h(G(u))G'(u), \quad B_{p,q} \sim \text{Beta}(p+1, q+1). \quad (52)$$

Let $N = p + q + 2$ and $\mu = (p+1)/N$. If p/N remains in a compact subinterval of $(0, 1)$ and $K \in C^4$ near μ , then

$$\mathbb{E}K(B_{p,q}) = K(\mu) + \frac{\mu(1-\mu)}{2(N+1)} K''(\mu) + O(N^{-2}). \quad (53)$$

In particular, when $p + q = n$ and $p/n \rightarrow \lambda \in (0, 1)$,

$$I_{p,q} \sim \sqrt{\frac{2\pi\lambda(1-\lambda)}{n}} \lambda^p (1-\lambda)^q G'(\lambda). \quad (54)$$

This is the ordinary, algebraic interior regime.

7.2 The balanced sequence and central flatness

The balanced case is exceptional. Define

$$K_m = (2m+1) \binom{2m}{m} I_{m,m} = \mathbb{E}G'(B_m), \quad B_m \sim \text{Beta}(m+1, m+1). \quad (55)$$

The center identity

$$F\left(\frac{1}{2} + h\right) = \frac{1}{2} + 2h - \text{sgn}(h)F(|h|) \quad (56)$$

implies

$$F'\left(\frac{1}{2} + h\right) = 2[1 - F(2|h|)]. \quad (57)$$

Because every derivative of F vanishes at 0, inverse differentiation gives

$$G'(1/2) = \frac{1}{2}, \quad G^{(r)}(1/2) = 0 \quad (r \geq 2). \quad (58)$$

Thus G' is flat, but not locally constant, at its minimum.

Theorem 7.1 (Monotone superalgebraic localization^N). *The sequence $(K_m)_{m \geq 0}$ is strictly decreasing and*

$$K_m \downarrow \frac{1}{2}. \quad (59)$$

Moreover, for every $A > 0$,

$$K_m - \frac{1}{2} = O_A(m^{-A}). \quad (60)$$

Proof. The function G' is symmetric about $1/2$. On $(1/2, 1)$, F' is strictly decreasing by (57), so $G'(u) = 1/F'(G(u))$ is strictly increasing. Hence G' increases with $|u - 1/2|$. The ratio of successive beta densities is a positive constant times $u(1 - u)$, which strictly decreases with $|u - 1/2|$. A one-crossing argument therefore gives $EG'(B_{m+1}) < EG'(B_m)$. Concentration gives the limit $G'(1/2) = 1/2$.

For any A , flatness (58) implies $|G'(u) - 1/2| \leq C_A |u - 1/2|^{2A}$ in a central neighborhood. The corresponding beta moment is $O(m^{-A})$, while the complement has exponentially small beta probability. This proves (60). $\square$

7.3 A logarithmic law and Lambert saddle

Superalgebraic decay conceals a precise logarithmic scale.

Theorem 7.2 (Balanced logarithmic asymptotic^N). *Assume the endpoint logarithm (16). Then*

$$\lim_{m \rightarrow \infty} \frac{\log(K_m - \frac{1}{2})}{(\log m)^2} = -\frac{1}{8 \log 2}. \quad (61)$$

The saddle in the central displacement $v = |u - 1/2| = e^{-t}$ is located at

$$t_m = \frac{1}{2} W(16m \log 2) + o(1). \quad (62)$$

Proof at logarithmic precision. For $u = 1/2 + v$, write $G(u) = 1/2 + h$. Equations (56)–(57) imply $v = 2h - F(h) \sim 2h$ and

$$G'(1/2 + v) - \frac{1}{2} = \frac{F(2h)}{2(1 - F(2h))}.$$

Consequently,

$$\log\left(G'(1/2 + v) - \frac{1}{2}\right) = -\frac{(\log(1/v))^2}{2 \log 2} + O(\log(1/v) \log \log(1/v)). \quad (63)$$

The beta density contributes $(1 - 4v^2)^m = \exp(-4mv^2 + O(mv^4))$. At logarithmic precision the relevant exponent is therefore

$$\Psi_m(t) = 4me^{-2t} + \frac{t^2}{2 \log 2}.$$

Its stationary equation is $8me^{-2t} = t/\log 2$, or $(2t)e^{2t} = 16m \log 2$, giving (62). Since $t_m \sim \frac{1}{2} \log m$,

$$\Psi_m(t_m) = \frac{t_m^2 + t_m}{2 \log 2} = \frac{(\log m)^2}{8 \log 2} + o((\log m)^2).$$

Standard upper and lower Laplace bounds on windows around the saddle prove (61). $\square$

Conjecture 7.3 (Full Lambert-periodic expansion^C). *Let $w_m = W(16m \log 2)$. The quantity $K_m - 1/2$ has a full expansion of the form*

$$K_m - \frac{1}{2} \sim \mathcal{A}(\vartheta_m) m^\delta w_m^\gamma \exp[-(w_m^2 + 2w_m)/(8 \log 2)] \left(1 + \sum_{j \geq 1} \frac{P_j(w_m)}{w_m^j} \right), \quad (64)$$

where $\mathcal{A}$ is a nonconstant bounded periodic factor inherited from the endpoint phase, ϑ_m is an affine function of $\log_2 w_m$, and the algebraic prefactor and polynomials P_j are determined by the all-orders endpoint expansion of F .

The exact periodic phase should be extracted from the corpus's Lambert-phase normal form rather than guessed from moderate values of m ; the convergence to the scale in (61) is extremely slow.

8 Powers of Rvachev's up function

Because $\text{up}(x) = 1 - F(x)$ on $[0, 1]$ and up is even, powers of up reduce to inverse-beta moments.

Theorem 8.1 (Power and weighted-power formulas^N). *For $\Re \alpha > -1$ and $\Re q > 0$,*

$$\boxed{\int_{-1}^1 |x|^\alpha \text{up}(x)^q dx = \frac{2q}{\alpha + 1} \int_0^1 G(u)^{\alpha+1} (1-u)^{q-1} du.} \quad (65)$$

If $q \in \mathbb{N}_{>0}$, then

$$\int_{-1}^1 |x|^\alpha \text{up}(x)^q dx = \frac{2}{\alpha + 1} \mathbb{E} Z_{1:q}^{\alpha+1}. \quad (66)$$

In particular,

$$\int_{-1}^1 \text{up}(x)^q dx = 2A_q = 2\mathbb{E} Z_{1:q}. \quad (67)$$

Proof. On $[0, 1]$, integrate $x^\alpha (1 - F(x))^q$ by parts. The boundary terms vanish and substitution $u = F(x)$ gives

$$\int_0^1 x^\alpha (1 - F(x))^q dx = \frac{q}{\alpha + 1} \int_0^1 G(u)^{\alpha+1} (1-u)^{q-1} du.$$

Double by evenness. For integer q , the beta variable $\text{Beta}(1, q)$ is $F(Z_{1:q})$. $\square$

The incomplete one-sided primitive is

$$\int_0^x t^\alpha \text{up}(t)^q dt = \frac{x^{\alpha+1} (1 - F(x))^q}{\alpha + 1} + \frac{q}{\alpha + 1} Q_{\alpha+1}(1, q; F(x)). \quad (68)$$

For $q = 1$,

$$\int_{-1}^1 |x|^\alpha \text{up}(x) dx = \frac{2d_{\alpha+1}}{\alpha + 1}. \quad (69)$$

8.1 Coupling to the derivative of F

The derivative removes all distribution-specific complexity:

$$\int_0^1 F(x)^p (1 - F(x))^q F'(x) dx = B(p + 1, q + 1), \quad (70)$$

$$\int_0^1 F(x)^p (1 - F(x))^q \operatorname{up}(2x - 1) dx = \frac{1}{2} B(p + 1, q + 1), \quad (71)$$

$$\int_0^1 x^r F(x)^p (1 - F(x))^q F'(x) dx = Q_r(p + 1, q + 1). \quad (72)$$

These identities are exact for complex parameters throughout their domains of absolute convergence.

9 A transform dictionary

9.1 Transforms of the inverse Fabius function

The quantile substitution gives the master identity

$$\int_0^1 \Psi(G(u)) du = \mathbb{E}\Psi(X). \quad (73)$$

Therefore

$$\int_0^1 e^{zG(u)} du = M_X(z) = \prod_{j \geq 1} \frac{e^{z/2^j} - 1}{z/2^j}, \quad (74)$$

$$\int_0^1 e^{-2\pi i \xi G(u)} du = e^{-\pi i \xi} \Phi(\xi/2), \quad (75)$$

$$\int_0^1 (z - G(u))^{-\nu} du = \mathbb{E}(z - X)^{-\nu}, \quad (76)$$

$$\int_0^1 G(u)^r (1 - G(u))^s du = \mathbb{E}X^r (1 - X)^s. \quad (77)$$

When $s \in \mathbb{N}$, the last expression is the finite rational-moment sum

$$\sum_{k=0}^s (-1)^k \binom{s}{k} d_{r+k}. \quad (78)$$

9.2 Linear transforms of F

Integration by parts gives

$$\int_0^1 e^{-sx} F(x) dx = \frac{M_X(-s) - e^{-s}}{s}, \quad (79)$$

$$\widehat{F\mathbf{1}_{[0,1]}}(\xi) = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}, \quad (80)$$

$$\int_0^1 x^{s-1} F(x) dx = \frac{1 - d_s}{s}, \quad (81)$$

$$\int_0^1 x^{s-1}(1-F(x)) dx = \frac{d_s}{s}. \quad (82)$$

The removable values at $s = 0$ or $\xi = 0$ are taken by continuity.

For G , one obtains the nonlinear Laplace dual

$$\int_0^1 e^{-su} G(u) du = \frac{\int_0^1 e^{-sF(x)} dx - e^{-s}}{s} = \frac{\sum_{n \geq 0} \frac{(-s)^n}{n!} A_n - e^{-s}}{s}. \quad (83)$$

Thus the Laplace transform of the quantile is encoded by the complete mixed-power edge sequence (A_n) .

9.3 Spacing transforms

For $p, q > 0$, the spacing theorem yields an entire transform package. With $N = p + q$,

$$\begin{aligned} \mathcal{M}_{p,q}(s) &:= \int_0^1 x^{s-1} F(x)^p (1-F(x))^q dx \\ &= \frac{\mathbb{E}(Z_{p+1:N}^s - Z_{p:N}^s)}{s \binom{N}{p}}, \end{aligned} \quad (84)$$

$$\begin{aligned} \mathcal{L}_{p,q}(s) &:= \int_0^1 e^{-sx} F(x)^p (1-F(x))^q dx \\ &= \frac{\mathbb{E}(e^{-sZ_{p:N}} - e^{-sZ_{p+1:N}})}{s \binom{N}{p}}, \end{aligned} \quad (85)$$

$$\begin{aligned} \mathcal{C}_{p,q}(z) &:= \int_0^1 \frac{F(x)^p (1-F(x))^q}{z-x} dx \\ &= \frac{1}{\binom{N}{p}} \mathbb{E} \log \frac{z - Z_{p:N}}{z - Z_{p+1:N}}. \end{aligned} \quad (86)$$

More generally, for $\nu \neq 1$,

$$\int_0^1 \frac{F(x)^p (1-F(x))^q}{(z-x)^\nu} dx = \frac{\mathbb{E}[(z - Z_{p+1:N})^{1-\nu} - (z - Z_{p:N})^{1-\nu}]}{(\nu-1) \binom{N}{p}}. \quad (87)$$

All branch choices are understood consistently on a simply connected domain avoiding $[0, 1]$.

10 The Cauchy–Stieltjes hierarchy of the up-law

For $z \in \mathbb{C} \setminus [-1, 1]$ and $\Re \nu > 0$, define

$$S_\nu(z) = \int_{-1}^1 \frac{\text{up}(y)}{(z-y)^\nu} dy = \mathbb{E}(z - Y)^{-\nu}. \quad (88)$$

10.1 Laplace–sinh representation

For $\Re z > 1$,

$$S_\nu(z) = \frac{1}{\Gamma(\nu)} \int_0^\infty t^{\nu-1} e^{-zt} \prod_{j \geq 1} \frac{\sinh(t/2^j)}{t/2^j} dt. \quad (89)$$

Combining (14) with Watson expansion yields a complete large- z Bell–Bernoulli asymptotic; because the support is compact, it is in fact convergent for $|z| > 1$:

$$S_\nu(z) = z^{-\nu} \sum_{m \geq 0} \frac{(\nu)_{2m}}{(2m)!} \frac{c_m}{z^{2m}}. \quad (90)$$

Every coefficient is rational.

10.2 Dyadic functional equations

Theorem 10.1 (Dyadic Stieltjes hierarchy^N). *For $\nu \neq 1$,*

$$S_\nu(z) = \frac{2^{\nu-1}}{\nu-1} [S_{\nu-1}(2z-1) - S_{\nu-1}(2z+1)]. \quad (91)$$

At $\nu = 1$,

$$S_1(z) = \mathbb{E} \log \frac{2z+1-Y}{2z-1-Y}. \quad (92)$$

Furthermore,

$$S_\nu^{(m)}(z) = (-1)^m (\nu)_m S_{\nu+m}(z), \quad (93)$$

$$S_1'(z) = 2 [S_1(2z+1) - S_1(2z-1)], \quad (94)$$

$$S_1(-z) = -S_1(z). \quad (95)$$

Proof. The self-similarity $Y = (V + Y')/2$, with $V \sim \text{Unif}[-1, 1]$ independent of Y' , gives

$$S_\nu(z) = 2^{\nu-1} \mathbb{E} \int_{-1}^1 (2z - Y' - v)^{-\nu} dv.$$

The elementary integral in v gives (91); its logarithmic limit is (92). Differentiation proves (93), and the case $\nu = 2$ of (91) gives (94). Symmetry of Y gives (95). $\square$

10.3 Boundary values and a Hilbert-transform refinement law

For $-1 < x < 1$, the upper boundary value satisfies

$$S_{1,+}(x) = \text{PV} \int_{-1}^1 \frac{\text{up}(t)}{x-t} dt - \pi i \text{up}(x). \quad (96)$$

Taking imaginary and real parts of (94) yields respectively the known dyadic refinement of up and a parallel functional equation for its Hilbert transform. Thus the Cauchy transform packages the density and its harmonic conjugate into a single dyadic object.

10.4 Endpoint derivative growth

At the right endpoint,

$$\frac{(-1)^n}{n!} S_1^{(n)}(1) = 2^{-n-1} \mathbb{E} X^{-n-1}. \quad (97)$$

The endpoint law (16), by a one-dimensional Laplace principle, implies

$$\log \mathbb{E} X^{-s} = \frac{\log 2}{2} s^2 + O(s \log s). \quad (98)$$

Hence

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \log \left| \frac{S_1^{(n)}(1)}{n!} \right| = \frac{\log 2}{2}. \quad (99)$$

The one-sided derivatives at 1 all exist, but the Taylor series there has radius zero. This is a transform-level signature of the flat endpoint of up.

11 Generalized resolvent duality for F and G

For $z \notin [0, 1]$, define

$$C_{F,v}(z) = \int_0^1 \frac{F(x)}{(z-x)^v} dx, \quad (100)$$

$$C_{G,v}(z) = \int_0^1 \frac{G(u)}{(z-u)^v} du. \quad (101)$$

Theorem 11.1 (CDF/quantile resolvent duality^N). For $v \neq 1$,

$$C_{F,v}(z) = \frac{(z-1)^{1-v} - \mathbb{E}(z-X)^{1-v}}{v-1}, \quad (102)$$

$$C_{G,v}(z) = \frac{(z-1)^{1-v} - \int_0^1 (z-F(x))^{1-v} dx}{v-1}. \quad (103)$$

At $v = 1$,

$$C_F(z) = \mathbb{E} \log(z-X) - \log(z-1), \quad (104)$$

$$C_G(z) = \int_0^1 \log(z-F(x)) dx - \log(z-1). \quad (105)$$

Both satisfy the same reflection law

$$C_F(z) - C_F(1-z) = C_G(z) - C_G(1-z) = \log \frac{z}{z-1}. \quad (106)$$

Proof. Use the primitive $(z-x)^{1-v}/(v-1)$ and integrate by parts. In the quantile case, $\int H(u) dG(u) = \int H(F(x)) dx$. The logarithmic formulas are the limits as $v \rightarrow 1$. Reflection follows by substituting $x \mapsto 1-x$ and using the symmetries of F and G . $\square$

Let

$$T(z) = \mathbb{E}(z-X)^{-1} = 2S_1(2z-1). \quad (107)$$

The affine self-similarity $X = (U + X')/2$ yields

$$T(z) = 2\mathbb{E} \log \frac{2z-X}{2z-1-X}, \quad (108)$$

$$T'(z) = 4[T(2z) - T(2z-1)]. \quad (109)$$

These formulas form a nonlinear/linear pair: (108) is a logarithmic potential identity, while (109) is a linear dyadic functional-differential equation.

12 Sinc-product energies and the constant $\int_0^1 F^2$

Set

$$A_2 = \int_0^1 F(x)^2 dx. \quad (110)$$

Numerically,

$$A_2 = 0.404436767983985281465621491145 \dots \quad (111)$$

No familiar elementary or classical-special-function closed form is known in the audited corpus. There is, however, an exact sinc-product energy form.

Theorem 12.1 (Sinc energy and complementary Gini sum rule^N). *With Φ as in (11),*

$$A_2 = \int_0^\infty \Phi(\xi)^2 d\xi, \quad (112)$$

$$\mathbb{E}|X - X'| = 1 - 2A_2 = \frac{1}{2\pi^2} \int_0^\infty \frac{1 - \Phi(\xi)^2}{\xi^2} d\xi, \quad (113)$$

$$\boxed{\int_0^\infty \Phi(\xi)^2 d\xi + \frac{1}{4\pi^2} \int_0^\infty \frac{1 - \Phi(\xi)^2}{\xi^2} d\xi = \frac{1}{2}.} \quad (114)$$

Proof. Since $\text{up}(x) = 1 - F(x)$ on $[0, 1]$, symmetry and Parseval give

$$\int_{\mathbb{R}} \text{up}(x)^2 dx = 2A_2 = \int_{\mathbb{R}} \Phi(\xi)^2 d\xi,$$

which is (112). The standard cycles-normalized fractional-energy identity at exponent one is

$$\mathbb{E}|Z| = \frac{1}{\pi^2} \int_0^\infty \frac{1 - \Re \varphi_Z(\xi)}{\xi^2} d\xi.$$

For $Z = X - X'$, $|\varphi_X(\xi)|^2 = \Phi(\xi/2)^2$, yielding (113) after a change of variables. Finally, $\mathbb{E}|X - X'| = 2 \int_0^1 F(1 - F) = 1 - 2A_2$. $\square$

The integrand in (113) is regular at the origin:

$$\lim_{\xi \rightarrow 0} \frac{1 - \Phi(\xi)^2}{\xi^2} = \frac{4\pi^2}{9}, \quad (115)$$

because $\text{Var}(Y) = 1/9$.

For every integer $q \geq 2$, Fourier convolution gives the higher-power representation

$$A_q = \frac{1}{2} \int_{\mathbb{R}^{q-1}} \Phi(\xi_1) \cdots \Phi(\xi_{q-1}) \Phi\left(-\sum_{j=1}^{q-1} \xi_j\right) d\xi_1 \cdots d\xi_{q-1}. \quad (116)$$

For $0 < \alpha < 2$, the full fractional energy is

$$\mathbb{E}|X - X'|^\alpha = \frac{2^{1-\alpha} \Gamma(1 + \alpha) \sin(\pi\alpha/2)}{\pi^{1+\alpha}} \int_0^\infty \frac{1 - \Phi(\xi/2)^2}{\xi^{1+\alpha}} d\xi. \quad (117)$$

Conjecture 12.2 (Arithmetic opacity of the energy constant^C). *The number A_2 is transcendental. More conservatively, it does not belong to the field generated over $\overline{\mathbb{Q}}$ by finitely many values of Gamma, zeta, polylogarithm, and elementary functions at algebraic arguments.*

The conjecture is deliberately separated from the exact formulas (112) and (114); those formulas may instead suggest a new class of “dyadic sinc periods.”

13 General derivatives and antiderivatives of arbitrary order

13.1 Finite Thue–Morse derivative formulas

Let $s_2(j)$ be the binary digit sum and $T_n = n(n+1)/2$. Iterating (9) gives the baseline finite formulas

$$\text{up}^{(n)}(x) = 2^{T_n} \sum_{j=0}^{2^n-1} (-1)^{n+s_2(j)} \text{up}(2^n x + 2j + 1 - 2^n), \quad (118)$$

$$F^{(n)}(x) = 2^{T_n} \sum_{j=0}^{2^{n-1}-1} (-1)^{n-1+s_2(j)} \text{up}(2^n x + 2j + 1 - 2^n), \quad n \geq 1. \quad (119)$$

For a mixed power $W_{p,q}(x) = F(x)^p(1-F(x))^q$, Faà di Bruno gives the complete derivative formula

$$W_{p,q}^{(n)}(x) = \sum_{k=0}^n \frac{d^k}{du^k} [u^p(1-u)^q] \Big|_{u=F(x)} B_{n,k}(F'(x), \dots, F^{(n-k+1)}(x)), \quad (120)$$

where $B_{n,k}$ is a partial exponential Bell polynomial and

$$\frac{d^k}{du^k} [u^p(1-u)^q] = \sum_{r=0}^k \binom{k}{r} p^r (-1)^{k-r} q^{k-r} u^{p-r} (1-u)^{q-k+r}. \quad (121)$$

Substituting (119) makes (120) a finite expression in Thue–Morse translates of up .

13.2 Integer and fractional primitives of up

The normalized n -fold primitive from $-\infty$ is

$$U_n(x) = \frac{1}{(n-1)!} \int_{-\infty}^x (x-t)^{n-1} \text{up}(t) dt. \quad (122)$$

For $x \leq 1$,

$$U_n(x) = 2^{\binom{n}{2}} F\left(\frac{x+1}{2^n}\right), \quad (123)$$

whereas for $x \geq 1$, compact support gives the moment polynomial

$$U_n(x) = \sum_{k=0}^{n-1} \frac{(-1)^k \mu_k}{k!(n-1-k)!} x^{n-1-k}, \quad \mu_k = \mathbb{E}Y^k. \quad (124)$$

The odd μ_k vanish, and the even ones are the Bell–Bernoulli moments in (14).

13.3 Fractional primitives of F and G

The baseline probabilistic form for F is

$$I^\nu F(x) = \frac{1}{\Gamma(\nu+1)} \mathbb{E}(x-X)_+^\nu. \quad (125)$$

For the inverse,

$$I^\nu G(y) = \frac{1}{\Gamma(\nu+1)} \int_0^{G(y)} (y-F(t))^\nu dt. \quad (126)$$

The integer derivatives of the inverse are compactly expressed by the inverse operator

$$G^{(n)}(u) = \left[\left(\frac{1}{F'(x)} \frac{d}{dx} \right)^{n-1} \frac{1}{F'(x)} \right]_{x=G(u)}. \quad (127)$$

Equivalently, one may expand (127) using Bell polynomials in the ratios $F^{(k)}/F'$. Equations (120), (27), (123), and (127) together provide a uniform answer to the “ n -th derivative/antiderivative” part of the problem.

14 Finite Thue–Morse splines and certified series approximations

Let

$$X_M = \sum_{j=1}^M \frac{U_j}{2^j}, \quad F_M(x) = P(X_M \leq x). \quad (128)$$

The generalized Irwin–Hall formula is

$$F_M(x) = \frac{2^{M(M+1)/2}}{M!} \sum_{k=0}^{2^M-1} (-1)^{s_2(k)} \left(x - \frac{k}{2^M} \right)_+^M. \quad (129)$$

Thus F_M is a rational piecewise polynomial on the dyadic mesh. The tail coupling satisfies

$$0 \leq X - X_M \leq 2^{-M} \quad \text{almost surely.} \quad (130)$$

Theorem 14.1 (Certified rational approximation^N). *Let $p, q \in \mathbb{N}_{>0}$, $N = p + q$, and let H be Lipschitz on $[0, 1]$ with constant L . Define*

$$J = \int_0^1 H'(x) F(x)^p (1 - F(x))^q dx, \quad J_M = \int_0^1 H'(x) F_M(x)^p (1 - F_M(x))^q dx.$$

Then

$$|J - J_M| \leq \frac{2L}{\binom{N}{p}} 2^{-M}. \quad (131)$$

If H' is a polynomial with rational coefficients, then $J_M \in \mathbb{Q}$ and can be evaluated exactly by integrating finitely many dyadic polynomials.

Proof. Couple every observation with its truncation using the same uniforms. By (130), every corresponding order statistic differs by at most 2^{-M} . Apply (17) to F and F_M ; the difference of the two adjacent-gap increments of H is at most $2L2^{-M}$. Formula (129) proves rationality. $\square$

Consequently, every integer mixed moment has a rational sequence representation with a certified exponential error. This complements the more rapidly convergent but structurally specialized q -Richardson schemes in the corpus. For noninteger powers, one may combine (129) with the Abel series (44), or use the beta-quantile integral directly.

15 Numerical experiments

The archive contains `fabius_integral_transforms.py`. It constructs a uniform-grid fixed point for F , computes exact rational moments by (6), evaluates the sinc product, and checks independent sides of the principal identities. The complete output is in `numerical_results.txt`. Selected results are shown below.

Check	Absolute residual
$F(1/4) = 5/72$, grid 2^{17}	1.29×10^{-11}
Weighted spacing identity, $e^x F^2(1 - F)^3$	3.35×10^{-12}
Fractional spacing primitive, order 1.65	1.26×10^{-12}
Cauchy equation (94)	below 2.3×10^{-16}
Sinc/Gini sum rule (114)	8.1×10^{-15}

The two independent estimates

$$\int_0^1 F(x)^2 dx \approx 0.4044367679787557$$

from the grid and

$$\int_0^{64} \Phi(\xi)^2 d\xi \approx 0.4044367679839853$$

from oscillatory quadrature agree to the expected grid accuracy. The normalized balanced values begin

$$K_1 \approx 0.5733793921, \quad K_{10} \approx 0.5030089983, \quad K_{100} \approx 0.5000147741, \quad K_{1000} \approx 0.50000000757,$$

consistent with strict decrease and superalgebraic localization. They are far from the ultimate logarithmic regime in (61), as the Lambert saddle moves only logarithmically.

16 Conjectures and frontier directions

16.1 Analytic continuation and non-holonomicity

Conjecture 16.1 (No local crossing of the Cauchy cut^C). *The Cauchy transform S_1 admits no analytic continuation through any nonempty open subinterval of $[-1, 1]$. Equivalently, the density up is not real-analytic on any nonempty open interval.*

A proof would imply that S_1 is not D-finite over $\mathbb{C}(z)$, since a D-finite function has only finitely many singular points in the finite plane. The dyadic equation (94), together with the nowhere-analytic structure of F , is the natural route.

16.2 The mixed-power resolvent as a new special function

The function R in (45) is a Stieltjes-type transform of the symmetric random variable $F(U)$, where $U \sim \text{Unif}[0, 1]$. It obeys the involutive Möbius law (47), but no closed form is known.

Problem 16.2. *Determine the maximal analytic continuation of R , its boundary values on $[1, \infty)$, and whether its Möbius functional equation plus positivity characterizes the Fabius pushforward measure.*

Problem 16.3. *Develop Padé and continued-fraction approximants to R from the exact Hausdorff moments A_n , and compare their pole distributions with the q -Richardson accelerants already present in the corpus.*

16.3 Balanced Lambert tomography

The full conjecture theorem 7.3 offers a new way to probe the periodic endpoint correction: balanced beta localization samples the flat central germ of G' , which is itself a reparameterized copy of the endpoint germ of F . Numerically fitting K_m after removing the Lambert saddle may recover the endpoint phase without evaluating F at extremely small arguments.

Problem 16.4. *Derive the first two explicit amplitude and phase terms in (64), and design a confluent q -Richardson scheme in the variable $w_m = W(16m \log 2)$ that preserves the periodic phase rather than averaging it away.*

16.4 Dyadic sinc periods

The constants

$$\mathcal{P}_q = \int_{\mathbb{R}^{q-1}} \Phi(\xi_1) \cdots \Phi(\xi_{q-1}) \Phi\left(-\sum \xi_j\right) d\xi = 2A_q \quad (132)$$

form a natural period family attached to the dyadic sinc product.

Problem 16.5. *Find recurrences, functional equations, or arithmetic obstructions for $\mathcal{P}_q$. Determine whether the complementary energy identity (114) is the first member of a hierarchy involving fractional Riesz energies and higher convolution powers.*

16.5 Formalization roadmap

The most Lean-friendly new results are:

1. the random-interval identity and adjacent-spacing theorem, using finite exchangeability and order statistics;
2. the Riemann–Liouville spacing formula, by Tonelli and an elementary truncated-power integral;
3. beta–quantile Pascal/reflection identities;
4. the Cauchy hierarchy, from the affine fixed-point law and integration of a uniform variable;
5. the sinc-energy identity, from Parseval and the standard Fourier formula for $|x|$;
6. the certified finite-spline bound, from the deterministic tail coupling.

The balanced logarithmic asymptotic requires the endpoint theorem plus a quantitative Laplace principle and should follow only after the elementary transform layer is formalized.

17 Compact formula compendium

For convenient reference, the central formulas are collected here.

Object	Formula
Weighted mixed integral	$\int h F^p (1 - F)^q = \binom{p+q}{p}^{-1} \mathbb{E}[H(Z_{p+1}) - H(Z_p)].$
Fractional mixed primitive	$I^\nu [F^p (1 - F)^q](x) = \frac{\mathbb{E}[(x - Z_p)_+^\nu - (x - Z_{p+1})_+^\nu]}{\Gamma(\nu + 1) \binom{p+q}{p}}.$

Object	Formula
Incomplete monomial primitive	Equation (32) in terms of truncated Q_{m+1} .
Beta-quantile reflection	$Q_1(a, b) + Q_1(b, a) = B(a, b)$.
Mixed-power recurrence	$I_{p,q} = I_{p+1,q} + I_{p,q+1} = (-1)^q \Delta^q A_p$.
Bivariate generating function	$\sum I_{p,q} z^p w^q = [zR(z) + wR(w)]/(z + w - zw)$.
Balanced normalization	$K_m = (2m+1) \binom{2m}{m} I_{m,m} \downarrow 1/2$, with $\log(K_m - 1/2) \sim -(\log m)^2/(8 \log 2)$.
Power of up	$\int_{-1}^1 x ^\alpha \text{up}(x)^q dx = \frac{2q}{\alpha+1} Q_{\alpha+1}(1, q)$.
Inverse Fourier transform	$\int_0^1 e^{-2\pi i \xi G(u)} du = e^{-\pi i \xi} \Phi(\xi/2)$.
up-Stieltjes hierarchy	$S_\nu(z) = \frac{2^{\nu-1}}{\nu-1} [S_{\nu-1}(2z-1) - S_{\nu-1}(2z+1)]$.
CDF resolvent	$C_{F,\nu}(z) = [(z-1)^{1-\nu} - \mathbb{E}(z-X)^{1-\nu}]/(\nu-1)$.
Sinc energy	$A_2 = \int_0^\infty \Phi^2$, and equation (114).
Finite spline	$F_M(x) = 2^{M(M+1)/2} M!^{-1} \sum_{k < 2^M} (-1)^{s_2(k)} (x - k/2^M)_+^M$.

18 Conclusion

The integration theory of the Fabius–Rvachev system becomes substantially more coherent when it is organized around three dual viewpoints:

1. *random intervals*: mixed CDF powers are occupation probabilities of gaps between two independent sample blocks;
2. *beta quantiles*: the substitution $u = F(x)$ turns mixed powers into beta localization against G' ;
3. *self-similar transforms*: the affine fixed-point laws of X and Y induce exact dyadic equations for Cauchy, Stieltjes, Laplace, and Fourier transforms.

These viewpoints cover indefinite integrals, all integer and fractional primitives, broad families of definite transforms, and nonlinear inverse-Fabius integrals without requiring a fictitious elementary closed form. They also expose new frontier phenomena: strict sign-regularity, balanced beyond-all-orders localization, endpoint derivative growth, and dyadic sinc periods. The resulting formulas are compatible with the repository's q-binomial, Lambert, Bell, Bernoulli, Thue–Morse, and sinc-product structures, but are not duplications of those earlier developments.

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